

JANET AKOTH KIRETA

+1 (423) 791 7793 | jantokireta@gmail.com | <https://www.linkedin.com/in/janet-kireta-08aba0254>

SUMMARY

Driven data analyst dedicated to helping organizations extract value from their data assets. Proficient at collecting, cleaning, and analyzing large datasets using various tools such as R, SAS, Python etc. Ability to communicate data-driven insights through compelling visualizations and presentations tailored to diverse audiences.

TECHNICAL COMPETENCE AND SKILLS

- **Programming Languages:** R | Python | SAS | SPSS | SQL | MATLAB
- **Markup Language:** LATEX
- **Other:** Windows | Microsoft Office (Word, PowerPoint, Excel, Outlook) | Zoom

EDUCATION

East Tennessee State University
Masters applied data science

August 2023 - May 2024

East Tennessee State University
Mathematical Science, M.S. (Statistics)

Aug 2021- May 2023

Southeastern Kenya University
Bachelor of Economics & Statistics

Jan 2011 – May 2015

RESEARCH AND SCHOLARLY ACTIVITIES

Machine Learning | Lung Cancer Classification | ETSU

Jan 2024 – May 2024

Evaluating and Improving ML Models for Predicting Lung Cancer through Feature Engineering and Classification Analysis

Advisor: Dr. Ahmad Al-Doulat- Professor in the Computer Science department

- Evaluated and improved ML models for predicting lung cancer by analyzing and preparing the dataset, engineering and selecting impactful features, developing and evaluating multiple classification models, optimizing model performance through tuning, and providing insights on model predictions and feature significance.

Chick-Fil-A.

Jan 2024 – May 2024

Evaluating Privacy Preserving Synthetic Data Generation

Advisor: Dr. Jason Joyner -Sr. Director Enterprise Data & Analytics (Chick-Fil-A) CFA

- Mitigated data sharing risks by using synthetic datasets, preventing sensitive information exposure, and implemented a codebase for synthetic data generation. Conducted a detailed comparative analysis of synthetic data methods to establish best practices and guide future projects.

Pfizer Inc.

Aug 2023 – May 2024

Build an in-silico predictor of the pressure-volume loop.

Advisor: Dr. Nessay Tania - Quantitative Systems Pharmacologist at Pfizer

- Led a team in collaboration with Pfizer, built an elastance model using provided data, and applied MATLAB for model validation, analysis, and reporting.

Artificial Intelligence | Network Intrusion Detection | ETSU Cybersecurity Research and Development Project

Aug 2023 – Dec 2023

Advisor: Dr. Ahmad Al-Doulat- Professor in the Computer Science department

- Investigated various effective machine learning algorithms for network intrusion detection.

Data Management | Data Manipulation with Excel | ETSU

Aug 2023 – Dec 2023

- Explored demographic data from multiple sources to extrapolate information about Farmers Markets in the United States by performing calculations and using formulas in Excel, joining two spreadsheets using lookups, and aggregating a summary of findings.

Unsupervised Dimension Reduction Techniques for Lung Cancer Diagnosis using Radiomics

Jan - May 2023

Advisor: Mostafa Zahed - Assistant Professor of Applied Statistics and Data Science, ETSU

- Applied unsupervised machine learning to feature engineer a large-scale CT dataset for lung cancer patients, contributing to the development of methods for preventing, detecting, diagnosing, treating, and ultimately curing cancer.

Sandia National Labs

Aug- Dec 2022

Surrogate Models for Nuclear Waste

Advisor: Dr. Bekah White – Scientific Machine Learning (Org. 1441) at Sandia

- Developed surrogate models for contaminant transport utilizing scientific machine learning and utilized Python, R, MATLAB, and Scikit-Learn for graphing and analytical tasks.

WORK EXPERIENCE

Research consultant | Statistical Consulting Lab (ETSU)

Aug 2023-May 2024

- Provided consultation services on research project methodologies, organized monthly workshops on survey design and statistical software like R, SPSS, Python, and SAS, advised on various machine learning techniques for data analysis, and assisted a professor in the smooth execution of tasks for partnership projects.

Research Assistant | Field Officer | Field Enumerator

May 2012 to Jul 2016

- Extensive research experience across multiple roles, focusing on data collection, analysis, and project coordination. My duties have included conducting one-on-one interviews, ensuring data accuracy, scheduling, and team coordination for various projects at Ideas42, Busara Center for Behavioral Economics, Center for Study of Adolescence, and OGRA Foundation. My work also involved facilitating and moderating focus groups, as well as handling data entry tasks, demonstrating my ability to uphold professional integrity and effectively contribute to research endeavors.

Industrial Experience

Planning Assistant | National Industrial Training Authority

Sep 2016- Aug 2021

- Played a pivotal role at the organization by taking part in strategic planning efforts, overseeing performance contract implementation and reporting, monitoring development projects, coordinating annual work plans, managing industrial training data analysis, and supporting trade test assessment activities.

PROFESSIONAL DEVELOPMENT

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| • SAS software application Southeast SAS User Group Conference | Oct 22 nd – 24 th 2023 |
| • Data Analysis using SPSS. | March 2021 |
| • Young Researcher Centre for Studies of Adolescence under Access, Service and Knowledge. | Sept 2014 |
| • Field Enumerator OGRA Foundation under the Healthy Start Program | Oct 2013 |
| • Computer Applications Kibuye Institute of Professional Studies. | April 2010 |

AWARDS

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| • Rotary District 7570 Skelton / Jones Scholarship 2023-2024 Scholar | Aug 2023 |
| • Appalachian Student Research Forum 1 st Place in the Biostatistics Category | April 2023 |

ACTIVITIES AND COMMUNITY OUTREACH

- Abstract Algebra Club | ETSU
- Python Club | ETSU
- Adventist Christian Fellowship | ETSU